analysis

April 19, 2026

[1]:
```
import nest_asyncio
nest_asyncio.apply()
```

[2]:
```
import pyshark
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression
from sklearn.metrics import r2_score
```

[3]:
```
pcap_path = "../captures/2.pcapng"
capture = pyshark.FileCapture(pcap_path, display_filter="icmp")
```

[4]:
```
packets = []

for pkt in capture:
    try:
        packets.append({
            "time": float(pkt.sniff_timestamp),
            "src": pkt.ip.src,
            "dst": pkt.ip.dst,
            "type": str(pkt.icmp.type),
            "seq": str(pkt.icmp.seq)
        })
    except Exception:
        pass

capture.close()

df = pd.DataFrame(packets)
df.head()
```

[4]:

	time	src	dst	type	seq
0	1.776503e+09	10.10.166.182	172.217.24.14	8	1
1	1.776503e+09	172.217.24.14	10.10.166.182	0	1
2	1.776503e+09	10.10.166.182	172.217.24.14	8	2
3	1.776503e+09	172.217.24.14	10.10.166.182	0	2
4	1.776503e+09	10.10.166.182	172.217.24.14	8	3

[5]:
```python
print("Shape:", df.shape)
print("\nICMP Types Count:")
print(df["type"].value_counts())
```

```
Shape: (129, 5)

ICMP Types Count:
type
8    75
0    54
Name: count, dtype: int64
```

[6]:
```python
requests = df[df["type"] == "8"].copy()    # Echo Request
replies = df[df["type"] == "0"].copy()     # Echo Reply

print("Requests:", len(requests))
print("Replies:", len(replies))
```

```
Requests: 75
Replies: 54
```

[14]:
```python
rtt_data = []

for _, req in requests.iterrows():
    match = replies[
        (replies["src"] == req["dst"]) &
        (replies["dst"] == req["src"]) &
        (replies["seq"] == req["seq"]) &
        (replies["time"] > req["time"])
    ]

    if not match.empty:
        rep = match.iloc[0]
        rtt = (rep["time"] - req["time"]) * 1000
        rtt_data.append({
            "destination": req["dst"],
            "rtt_ms": rtt
        })

rtt_df = pd.DataFrame(rtt_data)
rtt_df.head()
```

[14]:

	destination	rtt_ms
0	172.217.24.14	6.430387
1	172.217.24.14	9.650707
2	172.217.24.14	4.999399
3	172.217.24.14	5.030870
4	104.16.132.229	37.217617

[15]:
```python
avg_rtt = rtt_df.groupby("destination")["rtt_ms"].mean().reset_index()
avg_rtt
```

[15]:

	destination	rtt_ms
0	103.102.166.224	60.246992
1	104.16.132.229	39.125872
2	150.171.22.12	28.033352
3	150.171.28.10	27.694178
4	151.101.129.140	34.519482
5	17.253.144.10	31.894255
6	172.217.24.14	6.527841
7	172.64.154.211	25.846434
8	198.252.206.1	32.220936
9	20.207.73.82	36.958122
10	74.6.231.20	275.028324

[16]:
```python
avg_rtt["destination"]
```

[16]:
```
0     103.102.166.224
1      104.16.132.229
2       150.171.22.12
3       150.171.28.10
4     151.101.129.140
5       17.253.144.10
6       172.217.24.14
7      172.64.154.211
8       198.252.206.1
9        20.207.73.82
10        74.6.231.20
Name: destination, dtype: str
```

[17]:
```python
distance_map = {
    "103.102.166.224": 50,
    "104.16.132.229": 1800,
    "150.171.22.12": 2200,
    "150.171.28.10": 2200,
    "151.101.129.140": 2500,
    "17.253.144.10": 3000,
    "172.217.24.14": 1200,
    "172.64.154.211": 1800,
    "198.252.206.1": 3200,
    "20.207.73.82": 2500,
    "74.6.231.20": 8000
}

avg_rtt_df["distance_km"] = avg_rtt_df["destination_ip"].map(distance_map)
final_df = avg_rtt_df.dropna()
```

```
final_df
```

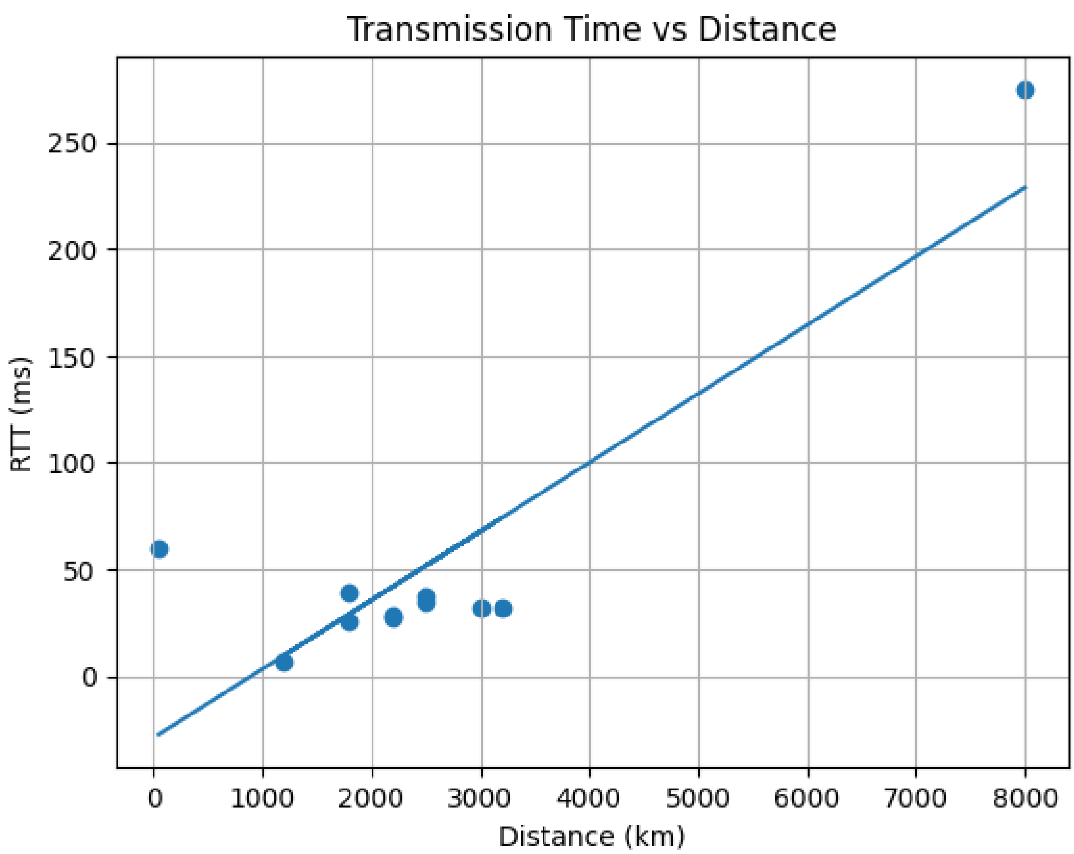
[17]:

	destination_ip	avg_rtt_ms	distance_km
0	103.102.166.224	60.246992	50
1	104.16.132.229	39.125872	1800
2	150.171.22.12	28.033352	2200
3	150.171.28.10	27.694178	2200
4	151.101.129.140	34.519482	2500
5	17.253.144.10	31.894255	3000
6	172.217.24.14	6.527841	1200
7	172.64.154.211	25.846434	1800
8	198.252.206.1	32.220936	3200
9	20.207.73.82	36.958122	2500
10	74.6.231.20	275.028324	8000

[18]:
```
X = final_df[["distance_km"]]
y = final_df["avg_rtt_ms"]

model = LinearRegression()
model.fit(X, y)

y_pred = model.predict(X)

print("Slope:", model.coef_[0])
print("Intercept:", model.intercept_)
print("R² Score:", r2_score(y, y_pred))
```

```
Slope: 0.03222331475041164
Intercept: -28.96886527979168
R² Score: 0.748489886075832
```

[19]:
```
plt.scatter(X, y)
plt.plot(X, y_pred)
plt.xlabel("Distance (km)")
plt.ylabel("RTT (ms)")
plt.title("Transmission Time vs Distance")
plt.grid()
plt.show()
```



[25]:
```
clean_df = final_df[final_df["avg_rtt_ms"] < 150]
```

[26]:
```
X_clean = clean_df[["distance_km"]]
y_clean = clean_df["avg_rtt_ms"]

model.fit(X_clean, y_clean)

print("Clean R²:", r2_score(y_clean, model.predict(X_clean)))
```

```
Clean R²: 0.08936566428814163
```